

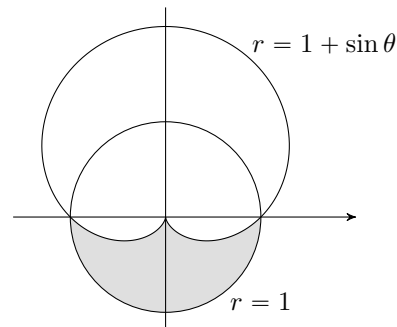
KUWAIT UNIVERSITY
Department of Mathematics

MATH 211: CALCULUS III

Fall 2018–19 • Second Exam • 20 November 2018

1. [5 pts] Which of the following holds for the function $f(x, y) = \sin x \cos y$ and the point $P(\frac{\pi}{2}, 0)$?
- (a) P is not a critical point of f . (b) f has a local minimum at P .
(c) f has a local maximum at P . (d) f has an absolute minimum at P .
(e) f has a saddle point at P .
2. [5 pts] Let E be the solid enclosed by the surfaces $x = y^2$, $x + z = 1$, and $z = 0$. Which of the following iterated integrals evaluates to the volume of E ?
- (a) $\int_{-1}^1 \int_0^{\sqrt{x}} (1-x) \, dy \, dx$ (b) $\int_{-1}^1 \int_{y^2}^1 (1-x) \, dx \, dy$ (c) $\int_0^1 \int_0^{\sqrt{x}} (1-x) \, dy \, dx$
(d) $\int_{-1}^1 \int_{-\sqrt{x}}^{\sqrt{x}} (1-x) \, dx \, dy$ (e) $\int_{-1}^1 \int_0^{y^2} (1-x) \, dx \, dy$
3. [5 pts] If $f(x, y)$ is continuous everywhere, which of the following equals $\int_0^2 \int_y^{\frac{y}{2}+1} f(x, y) \, dx \, dy$?
- (a) $\int_y^{\frac{y}{2}+1} \int_0^2 f(x, y) \, dy \, dx$
(b) $\int_0^1 \int_0^2 f(x, y) \, dy \, dx + \int_0^2 \int_0^{2x-2} f(x, y) \, dy \, dx$
(c) $\int_0^1 \int_0^x f(x, y) \, dy \, dx + \int_1^2 \int_{2x-2}^x f(x, y) \, dy \, dx$
(d) $\int_0^1 \int_1^x f(x, y) \, dy \, dx + \int_1^2 \int_x^{2x-2} f(x, y) \, dy \, dx$
(e) $\int_0^1 \int_1^y f(x, y) \, dy \, dx + \int_1^2 \int_y^{2y-2} f(x, y) \, dy \, dx$
4. [5 pts] Let E be the solid hemisphere that lies above the xy -plane and inside the sphere $x^2 + y^2 + z^2 = 1$. Which of the following is the value of $\iiint_E dV$?
- (a) 1 (b) $\frac{\pi}{3}$ (c) $\frac{1}{3}$ (d) $\frac{2\pi}{3}$ (e) π
5. [5 pts] which of the following is an equation for the plane $x + y + z = 1$?
- (a) $\rho = \frac{1}{\cos \phi + \sin \phi (\cos \theta + \sin \theta)}$ in spherical coordinates
(b) $\rho = \frac{1}{\cos \phi} + \frac{1}{\sin \phi \cos \theta} + \frac{1}{\sin \phi \sin \theta}$ in spherical coordinates
(c) $z = 1 + r(\cos \theta + \sin \theta)$ in cylindrical coordinates
(d) $\frac{z-1}{r(\cos \theta + \sin \theta)} = 1$ in cylindrical coordinates
(e) $(x + y + z)^2 = 1$ in Cartesian coordinates

6. [15 pts] Suppose that the temperature at each point on the sphere $x^2 + y^2 + z^2 = 21$ is given by the function $T(x, y, z) = 8x - 4y + 2z$. Find the hottest point on the sphere.
7. [15 pts] Use a double integral to find the area of the region that lies inside the circle $r = 1$ and outside the cardioid $r = 1 + \sin \theta$.



8. [15 pts] Let T denote the triangle in the xy -plane with vertices at $(0, 0)$, $(1, 0)$, and $(1, 2)$. Find the area of the part of the surface $z = x^2 + y$ that lies above T .
9. [5+5+5 pts] Let E be the solid that lies inside the sphere $x^2 + y^2 + z^2 = 8$ and above the cone $z = \sqrt{x^2 + y^2}$. Set up, but do not evaluate, an iterated triple integral for the volume of E in each of the following coordinate systems.
- Rectangular (Cartesian) coordinates.
 - Cylindrical coordinates.
 - Spherical coordinates.
10. [15 pts] A lamina occupies the region inside the circle $x^2 + y^2 = 3y$ and above the graph $y = |x|$. Find the center of mass of the lamina if the density function is $\rho(x, y) = \frac{1}{\sqrt{x^2 + y^2}}$.

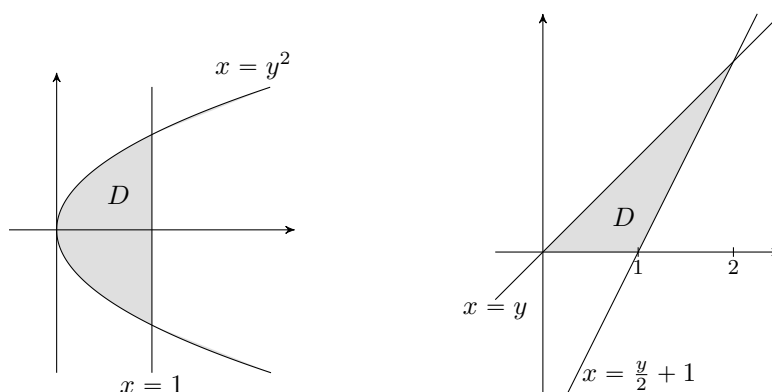
1. The correct choice is (c).

$f_x = \cos x \cos y$ and $f_y = -\sin x \sin y$ both vanish at P , thus P is a critical point of f . On the other hand, $f_{xx}(\frac{\pi}{2}, 0) = f_{yy}(\frac{\pi}{2}, 0) = -1$ and $f_{xy}(\frac{\pi}{2}, 0) = 0$ yield $D(\frac{\pi}{2}, 0) = 1 > 0$, and since $f_{xx} < 0$, the second derivatives test concludes f has a local maximum at P .

Since f is not a constant function, a local maximum of it cannot be a minimum (local or absolute) as well.

2. The correct choice is (b).

E is the solid under the graph $z = 1 - x$ and above the region D sketched on left below.



3. The correct choice is (c).

$$\int_0^2 \int_y^{\frac{y}{2}+1} f(x, y) \, dx \, dy = \iint_D f(x, y) \, dA \quad \text{where } D \text{ is the region sketched on right above.}$$

4. The correct choice is (d).

The triple integral represents the volume of a hemisphere with radius 1.

5. The correct choice is (a).

6. We must find the maximum values of T on the given sphere. Let $g(x, y, z) = x^2 + y^2 + z^2$. By the method of Lagrange multipliers, the maximum occurs at a solution of the system

$$\begin{cases} \nabla T = \lambda \nabla g \\ g = 21 \end{cases} \iff \begin{cases} 8 = \lambda(2x) \\ -4 = \lambda(2y) \\ 2 = \lambda(2z) \\ x^2 + y^2 + z^2 = 21. \end{cases}$$

Note that $\lambda \neq 0$ since otherwise, the first equation becomes $8 = 0$, a contradiction. Now dividing the first three equations by 2λ yields $x = \frac{4}{\lambda}$, $y = -\frac{2}{\lambda}$ and $z = \frac{1}{\lambda}$. Substituting these in the last equation we get

$$\frac{16}{\lambda^2} + \frac{4}{\lambda^2} + \frac{1}{\lambda^2} = 21 \iff \lambda^2 = 1 \iff \lambda = \pm 1.$$

So the system has two solutions $(4, -2, 1)$ and $(-4, 2, -1)$. Since $T(4, -2, 1) = 41$ and $T(-4, 2, -1) = -41$, we conclude that $(4, -2, 1)$ is the hottest point on the sphere.

7. Let D denote the given region. Then

$$\begin{aligned} A &= \iint_D dA = \int_{-\pi}^0 \int_{1+\sin\theta}^1 r \, dr \, d\theta = \int_{-\pi}^0 \frac{1}{2} [1 - (1 + \sin\theta)^2] d\theta \\ &= -\frac{1}{2} \int_{-\pi}^0 [2\sin\theta + \sin^2\theta] d\theta = -\frac{1}{2} \int_{-\pi}^0 \left[2\sin\theta + \frac{1 - \cos 2\theta}{2} \right] d\theta \\ &= \left[\cos\theta - \frac{\theta}{4} + \frac{1}{8} \sin 2\theta \right]_{-\pi}^0 = 2 - \frac{\pi}{4}. \end{aligned}$$

8. The triangle T can be represented as a Type I region by $0 \leq x \leq 1$ and $0 \leq y \leq 2x$. This gives

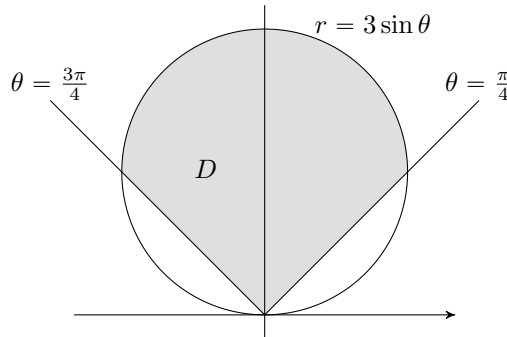
$$\begin{aligned} A(S) &= \iint_T \sqrt{1 + (\partial z / \partial x)^2 + (\partial z / \partial y)^2} \, dA = \int_0^1 \int_0^{2x} \sqrt{1 + 4x^2 + 1} \, dy \, dx \\ &= \frac{1}{4} \int_0^1 8x(4x^2 + 2)^{\frac{1}{2}} \, dx = \frac{1}{6} (4x^2 + 2)^{\frac{3}{2}} \Big|_0^1 = \frac{6^{\frac{3}{2}} - 2^{\frac{3}{2}}}{6} = \sqrt{6} - \frac{\sqrt{2}}{3}. \end{aligned}$$

9. (a) $\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{8-x^2-y^2}} dz \, dy \, dx$

(b) $\int_0^{2\pi} \int_0^2 \int_r^{\sqrt{8-r^2}} r \, dz \, dr \, d\theta$

(c) $\int_0^{2\pi} \int_0^{\frac{\pi}{4}} \int_0^{\sqrt{8}} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$

10. Let D denote the region occupied by the lamina (sketched below).



In polar coordinates, D is represented as $\frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4}$, $0 \leq r \leq 3\sin\theta$, and the density function is $\rho = 1/r$. By the line symmetry of the region D and the density function with respect to the y -axis, we have $\bar{x} = 0$. On the other hand,

$$m = \iint_D \rho(x, y) \, dA = \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \int_0^{3\sin\theta} dr \, d\theta = \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} 3\sin\theta \, d\theta = -3\cos\theta \Big|_{\frac{\pi}{4}}^{\frac{3\pi}{4}} = 3\sqrt{2},$$

and

$$\begin{aligned} \bar{y} &= \frac{1}{m} \iint_D y \rho(x, y) \, dA = \frac{1}{3\sqrt{2}} \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \int_0^{3\sin\theta} r \sin\theta \, dr \, d\theta = \frac{1}{3\sqrt{2}} \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{9}{2} \sin^3\theta \, d\theta \\ &= -\frac{3}{2\sqrt{2}} \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} (1 - \cos^2\theta)(-\sin\theta) \, d\theta = -\frac{3}{2\sqrt{2}} \left[\cos\theta - \frac{1}{3} \cos^3\theta \right]_{\frac{\pi}{4}}^{\frac{3\pi}{4}} = \frac{5}{4}. \end{aligned}$$